

Saranya Jothi Subramanian

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Education

Northeastern University

Master of Science - Software Engineering Systems

Boston, MA

Dec 2025

SASTRA University

Bachelor of Technology - Computer Science and Engineering

Tanjore, IN

Apr 2014

Technical Skills

Languages: Java, Python, SQL, C++

Frameworks: Executor, Spring Boot, Node.js, Express.js, RESTful APIs, JUnit, JMeter, Maven

AI/ML: LLaMA, LangGraph, RAG pipelines, FAISS, Hugging Face

Cloud & DevOps: AWS, Terraform, Docker, Kubernetes, Jenkins, Ansible, GitHub Actions, SonarQube, CI/CD

Databases: MySQL, PostgreSQL, MongoDB, Redis

Monitoring: Grafana, Splunk, New Relic, Sumo Logic

Architecture & Design: Microservices, Event-Driven Architecture (Kafka), Distributed Systems, Object Oriented Programming

Project Management: Agile-Scrum, Jira, Confluence, Kanban Board

Academic Projects

Meningioma Clinical Assistant – RAG-based GenAI System

Jul 2025 – present

- Built medical diagnosis support system using RAG pipeline with LLaMA 3 (8B) and FAISS, achieving 78-82% response accuracy on 400+ clinical queries and reducing retrieval latency by 60% (420ms to 170ms)
- Developed conversational AI prototype with LangGraph and ChatGPT API, enabling 6-8 turn interactions with 85-88% context retention across 1,200+ curated clinical documents
- Deployed Streamlit web application supporting 10-15 concurrent users, processing 60-80 PDF medical reports in <3 minutes with sub-3s page load time

Patient Management System

Apr 2025 – Jul 2025

- Built 3 microservices (Patient, Appointment, Billing) using Spring Boot and Kafka (3 partitions); load tested with JMeter achieving 200 concurrent requests with p95 latency of 380ms and p99 of 520ms at 450 req/sec throughput
- Implemented Redis caching (Redis Labs 30MB free tier) and Elasticsearch (Docker, 2 shards) for search; indexed 50K patient records with 120ms p95 search latency, achieving 48% cache hit rate and reducing MongoDB queries by 35%
- Containerized services with Docker Compose and deployed to AWS EC2 t2.medium for load testing; integrated Prometheus and Grafana for monitoring, achieving 99.2% uptime during 3-week testing period
- Automated CI/CD with GitHub Actions running Maven builds, JUnit tests (72% coverage), and Docker image creation; deployed to EC2 via SSH with 5-minute pipeline duration and 88% success rate across 45 builds

Professional Experience

LTIMindtree

Bangalore, IN

Specialist/Platform Software Engineer - ViacomCBS - CASL

Jul 2021 – Sep 2022

- Upgraded MongoDB driver from v3.8 to v4.1 over four sprints, achieving 45x performance boost and eliminating major bottlenecks
- Improved application performance by 25% through proactive MongoDB index monitoring and fine-tuning
- Developed custom named queries with Java and Spring, integrating into UI for intuitive query creation and data retrieval
- Optimized server-side workflows by identifying and resolving critical latencies, enhancing backend efficiency

Senior Software Developer - Apple Inc - Headcount Management Tool

May 2018 – Jun 2021

- Migrated application to Java 8, incorporating lambda expressions, functional interfaces, and streams for cleaner code and improved performance
- Implemented facade design pattern to streamline REST web service routing, simplifying architecture and improving maintainability
- Leveraged CompletableFuture for parallel database calls and ExecutorService for async email notifications, reducing DB hits and blocking operations while improving performance
- Authored Functional and Technical Specifications, ensuring clarity in requirements and technical details for client

Virtusa

Chennai, IN

Software Developer - Standard Chartered - iCDMS

May 2014 – May 2018

- Developed a utility tool using Java to validate and save documents for loan processing, reducing manual validation time by 60%
- Automated report generation for clients by creating PL/SQL Views, triggered to run automatically, improving process efficiency
- Developed a new search module to filter data and export search results as Excel files, enhancing data accessibility
- Parsed and validated Excel data, utilizing ActiveMQ to send data to upstream systems for object creation, optimizing data flow

Awards & Recognition

Client Appreciation Award - Standard Chartered (FY 2016 Q3)

Top Talent Award - Virtusa (FY 2015 Q4)

Best Find Award - Apple Inc (FY 2019 Q1)